Network Traffic Analysis using Wireshark

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1. Objective

The objective of this project is to capture and analyze network traffic using Wireshark in order to understand how data flows across a network and to identify basic communication patterns and to analyze the packet flow.

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2. Tools Used

- Wireshark (Network Protocol Analyzer)

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3. Methodology

The following steps were performed during the project:

1. Installed and configured Wireshark
2. Selected the active network interface (Wi-Fi)
3. Captured live network traffic
4. Generated traffic by browsing websites (Google, YouTube)
5. Stopped packet capture after a few minutes
6. Applied filters such as DNS and TCP
7. Analyzed packets and protocol behavior

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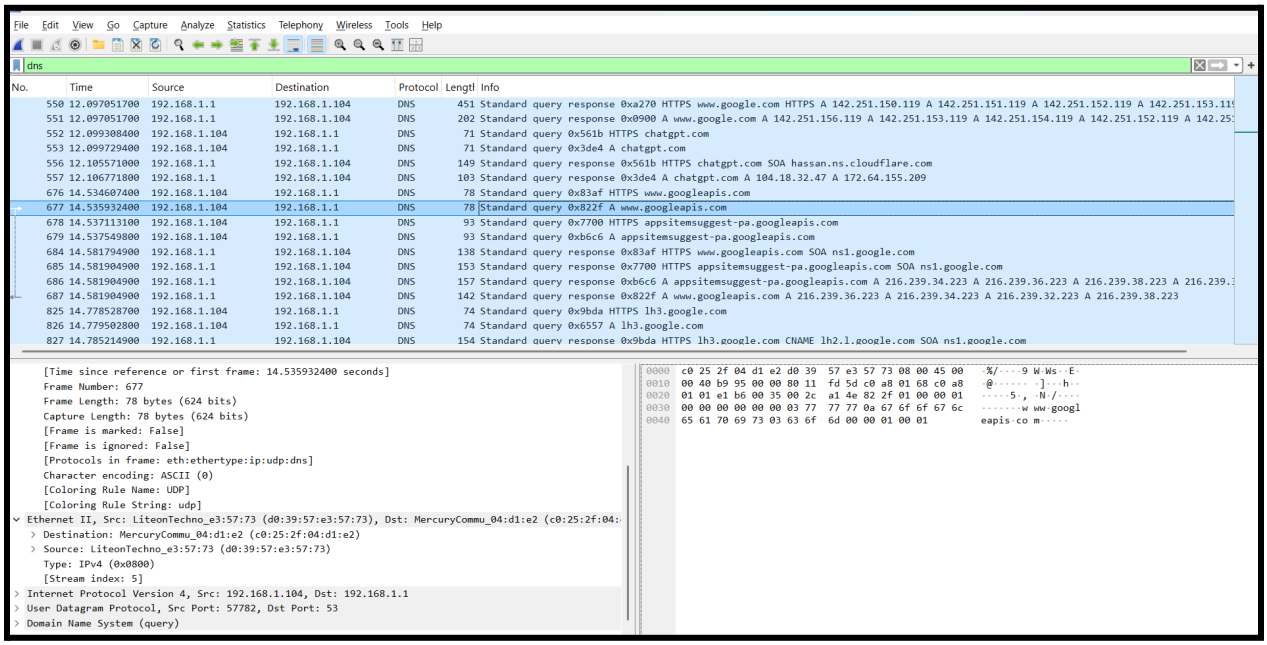
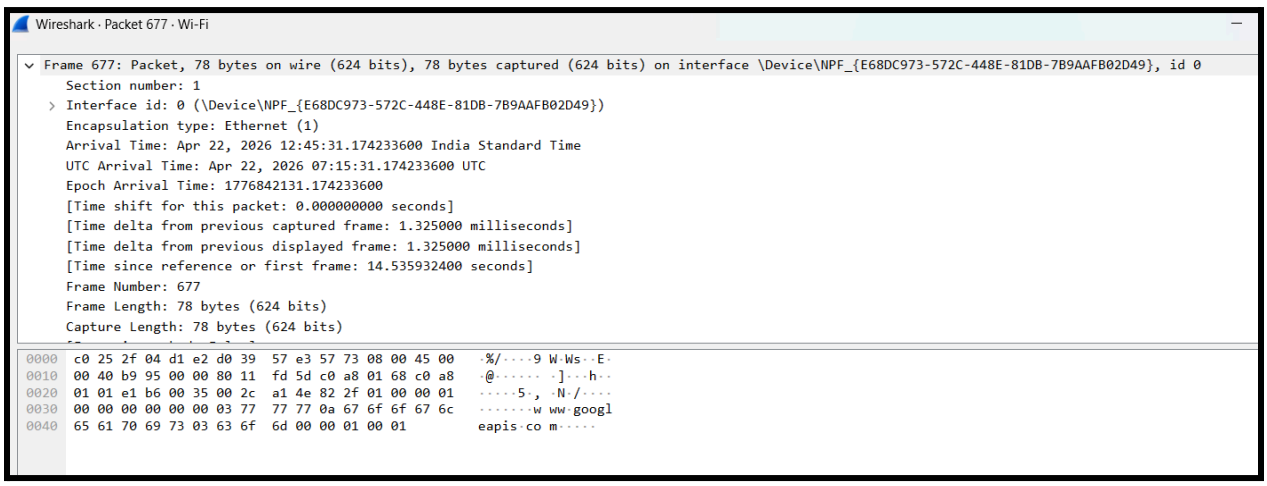
4. Analysis and Findings

4.1 DNS Analysis

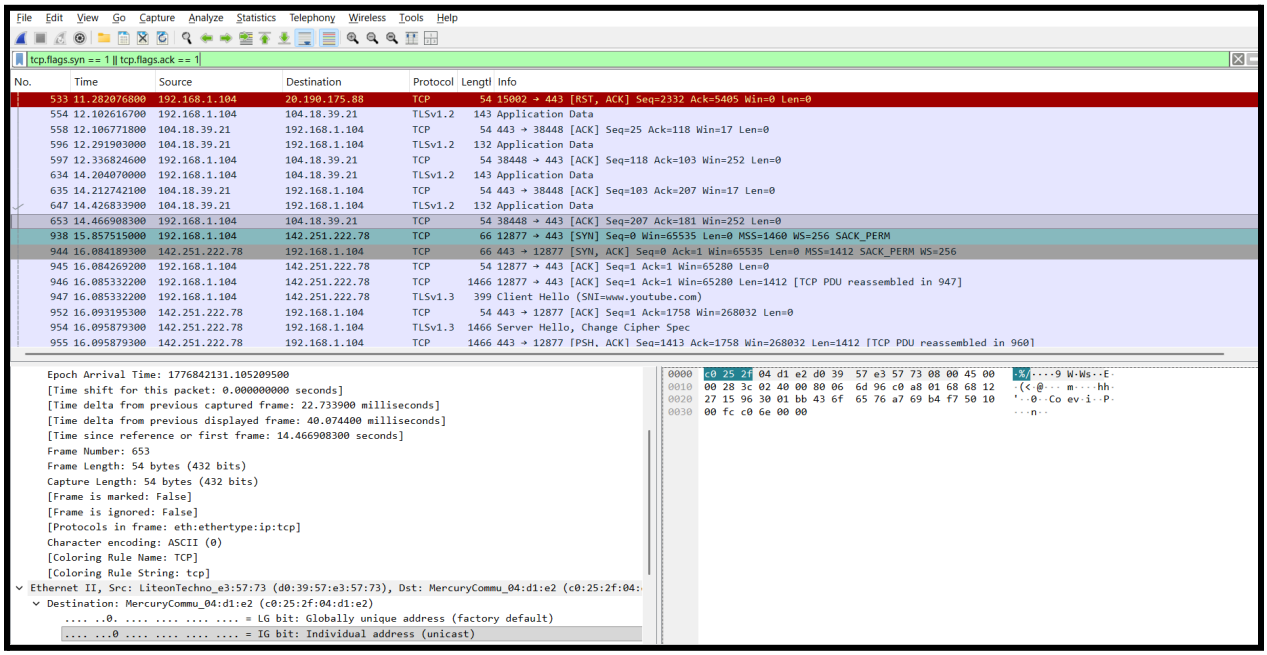
- Observed DNS queries for domain names such as google.com
- Identified corresponding IP addresses returned by DNS servers





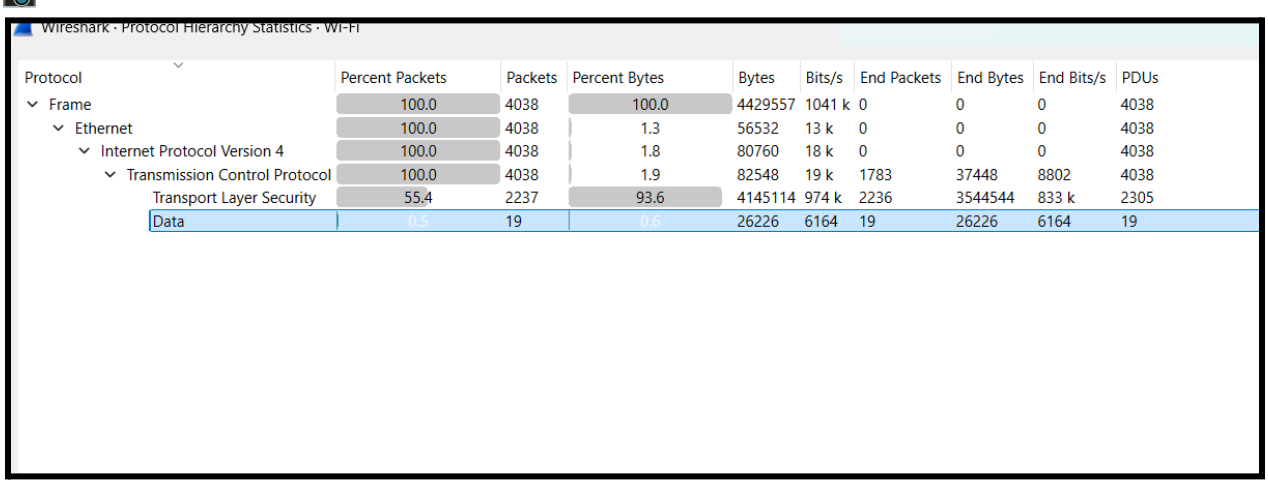
4.2 TCP Handshake Analysis

- Identified TCP 3-way handshake:
  - SYN
  - SYN-ACK
  - ACK
- Understood how connections are established between client and server



4.3 Protocol Hierarchy

- Majority of traffic consisted of TCP protocol
- DNS and HTTP traffic also observed



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	4038	100.0	4429557	1041 k	0	0	0	4038
Ethernet	100.0	4038	1.3	56532	13 k	0	0	0	4038
Internet Protocol Version 4	100.0	4038	1.8	80760	18 k	0	0	0	4038
Transmission Control Protocol	100.0	4038	1.9	82548	19 k	1783	37448	8802	4038
Transport Layer Security	55.4	2237	93.6	4145114	974 k	2236	3544544	833 k	2305
Data	0.5	19	0.6	26226	6164	19	26226	6164	19

4.4 IP Communication

- Observed communication between local system and external servers
- Identified source and destination IP addresses

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Ethernet · 13 | IPv4 · 70 | IPv6 | TCP · 53 | UDP · 141

Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
104.18.39.21	192.168.1.104	22	2 kB	31	22	100.00%	12	1 kB	10	954 bytes	7.582061900	27.697637	291 bits/s	275 bits/s
192.168.1.1	192.168.1.104	148	22 kB	23	148	100.00%	79	16 kB	69	6 kB	3.858097300	33.185571	3916 bits/s	1397 bits/s
192.168.1.1	224.0.0.252	5	230 bytes	18	5	100.00%	5	230 bytes	0	0 bytes	3.789825800	3.891694	472 bits/s	0 bits/s
192.168.1.1	239.255.255.250	4	1 kB	43	4	100.00%	4	1 kB	0	0 bytes	15.364907500	0.200906	53 kbps	0 bits/s
192.168.1.100	224.0.0.251	2	149 bytes	66	2	100.00%	2	149 bytes	0	0 bytes	32.259057700	4.500691	264 bits/s	0 bits/s
192.168.1.100	239.255.255.250	1	167 bytes	1	1	100.00%	1	167 bytes	0	0 bytes	0.001365400	0.000000		
192.168.1.101	224.0.0.251	1	289 bytes	2	1	100.00%	1	289 bytes	0	0 bytes	0.412468800	0.000000		
192.168.1.104	4.213.25.242	2	121 bytes	67	2	100.00%	1	55 bytes	1	66 bytes	33.609097100	0.315139	1396 bits/s	1675 bits/s
192.168.1.104	20.42.65.94	5	270 bytes	17	5	100.00%	3	162 bytes	2	108 bytes	3.752404700	0.563574	2299 bits/s	1533 bits/s
192.168.1.104	20.189.173.5	22	13 kB	60	22	100.00%	12	6 kB	10	7 kB	30.185194600	1.324976	33 kbps	44 kbps
192.168.1.104	20.190.146.37	80	69 kB	5	80	100.00%	33	17 kB	47	52 kB	3.243512300	1.072836	126 kbps	388 kbps
192.168.1.104	20.190.146.39	36	34 kB	25	36	100.00%	13	8 kB	23	26 kB	4.317167000	0.412713	160 kbps	504 kbps
192.168.1.104	20.190.175.88	18	9 kB	24	18	100.00%	10	3 kB	8	6 kB	4.264198600	7.017878	3286 bits/s	6666 bits/s
192.168.1.104	23.37.85.59	101	74 kB	65	101	100.00%	33	4 kB	68	69 kB	32.032330000	1.931206	18 kbps	287 kbps
192.168.1.104	35.227.194.49	3	162 bytes	20	3	100.00%	2	108 bytes	1	54 bytes	3.845557900	0.015516	55 kbps	27 kbps
192.168.1.104	38.106.231.208	35	17 kB	68	35	100.00%	16	4 kB	19	13 kB	34.185926200	1.124317	26 kbps	94 kbps
192.168.1.104	40.74.79.222	31	15 kB	27	31	100.00%	15	5 kB	16	10 kB	5.258753200	1.812712	20 kbps	43 kbps
192.168.1.104	40.100.7.128	48	24 kB	26	48	100.00%	21	6 kB	27	18 kB	5.246043800	1.537450	29 kbps	95 kbps
192.168.1.104	40.100.72.64	3	162 bytes	14	3	100.00%	2	108 bytes	1	54 bytes	3.751536100	0.106456	8116 bits/s	4058 bits/s
192.168.1.104	40.104.66.50	3	162 bytes	16	3	100.00%	2	108 bytes	1	54 bytes	3.752255200	0.105977	8152 bits/s	4076 bits/s
192.168.1.104	51.104.15.253	3	162 bytes	9	3	100.00%	2	108 bytes	1	54 bytes	3.686524200	0.171786	5029 bits/s	2514 bits/s
192.168.1.104	52.110.14.183	3	162 bytes	11	3	100.00%	2	108 bytes	1	54 bytes	3.688446500	0.036595	23 kbps	11 kbps
192.168.1.104	52.110.16.163	3	162 bytes	8	3	100.00%	2	108 bytes	1	54 bytes	3.685121200	0.038896	22 kbps	11 kbps
192.168.1.104	52.123.130.14	4	216 bytes	10	4	100.00%	2	108 bytes	2	108 bytes	3.687293200	0.170591	5064 bits/s	5064 bits/s
192.168.1.104	74.125.200.132	138	89 kB	53	138	100.00%	50	16 kB	88	73 kB	19.730614200	8.657932	14 kbps	67 kbps
192.168.1.104	103.88.222.200	706	767 kB	61	706	100.00%	114	14 kB	592	753 kB	30.942702100	0.246581	462 kbps	24 Mbps
192.168.1.104	103.88.223.14	547	703 kB	58	547	100.00%	68	33 kB	479	670 kB	28.482592500	0.369068	710 kbps	14 Mbps

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5. Conclusion

This project helped in understanding how network communication takes place at the packet level. Using Wireshark, different protocols such as DNS and TCP were analyzed, providing insights into real-world network behavior.

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6. Learning Outcomes

- Learned how to capture and analyze network packets
- Understood TCP/IP communication
- Gained hands-on experience with Wireshark
- Identified real-time network activity

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7. References

- Wireshark Official Documentation
- Basic Networking Concepts